Name: Janushiya P A

Reg No: 1924260368

Course Code: CSA0224

Course Name: C Programming

TASK 1 – Understand the Problem and Identify the Data

1. Aim

To analyse the electricity-billing problem and identify the inputs, outputs, constants, variables, rules, and assumptions required to develop a correct and reliable C program.

2. Problem Analysis

Grama Jyothi Rural Electricity Co-operative needs a C program to calculate household electricity bills. The program uses meter readings to calculate electricity consumption, applies the correct tariff, adds the fixed charge, and displays the total bill accurately to the paisa.

3. Inputs and Their Importance

- **Previous Reading:** The meter reading at the beginning of the billing period. It is required to calculate electricity consumption.
- **Current Reading:** The latest meter reading. Comparing it with the previous reading determines the units consumed.
- **Household Number:** Identifies each consumer and associates the bill with the correct household.
- **Household Category:** Determines the applicable tariff if different categories have different billing rates.

4. Outputs and Their Importance

- **Units Consumed:** The difference between the current and previous readings, adjusted for a valid meter rollover.
- **Energy Charge:** The amount calculated according to the electricity tariff slabs.
- **Fixed Charge:** The fixed amount charged for the billing period.
- **Total Bill:** The sum of the energy charge and fixed charge, including any other required charges.

5. Constants and Variables

Constants: Tariff rates, slab limits, fixed charge, and meter maximum value remain unchanged during a billing calculation. Named constants make the program easier to understand and maintain.

Variables: Meter readings, household number, category, units consumed, energy charge, and total bill may change for each household. Therefore, they must be stored in variables.

6. Negative, Fractional, and Large Values

Valid meter readings and units consumed cannot be negative. A smaller current reading may indicate meter rollover or an incorrect reading, so it must be checked.

Monetary calculations must be accurate to the paisa. Therefore, storing money as integer paise is preferable to using floating-point rupees. Large household counts and district-wide totals require data types with sufficient storage capacity.

7. Rules for a Correct Bill

1. Validate all inputs before calculation.
2. Calculate consumption correctly, including confirmed meter rollover.
3. Apply the correct tariff and household category.
4. Ensure the total equals the sum of all bill components.
5. Ensure the bill is not negative under the stated billing rules.
6. Display the amount accurately to two decimal places.

8. Assumptions

Each household has a unique number, and the co-operative provides the applicable tariff rates, slab limits, fixed charges, and category rules. If a 16-bit unsigned meter register is used, its values range from 0 to 65535. A decrease in reading is treated as rollover only when the meter's rollover behaviour is confirmed.

9. Alternative Concept: Text vs Integer Readings

A meter reading can be stored as a character string, such as "01530", instead of an integer.

Text preserves leading zeros, but the program must validate and convert it before performing calculations. For a reading containing *d* digits, conversion and validation require **O(d) time** and **O(d) storage**. A fixed-size integer requires **O(1) storage**, and comparison and arithmetic take constant time for a fixed-width type.

Therefore, integers are more suitable for billing calculations, while strings are useful for preserving the original reading format.

10. Conclusion / Result

The analysis identifies the data and rules required for accurate electricity billing. Using suitable data types, named constants, input validation, correct rollover handling, and integer paise helps produce reliable bills for individual households and large numbers of consumers.